UNIVERSITY OF MISSOURI-COLUMBIA
COLLEGE OF ENGINEERING
DEPARTMENT OF ELECTRICAL ENGINEERING AND COMPUTER SCIENCE

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APPENDIX A: A Case Study: Healthcare Document Management System (iCARE)

Overview

West Park (WPHC) is a public healthcare centre located in west Columbia City, Missouri, US that helps individuals manage difficult health challenges. WPHC delivery is highly regulated, it handles over 10,000 documents a day, each of which is touched by an average of ten people during its lifecycle. Some of the most important users and creators of these documents (doctors and nurses) have workflows that are deeply rooted in paper-dependent processes.

Jane Doe, who currently heads the new business intelligence division at WPHC, is looking for an efficient way to automate the paper-based processes while keeping costs to a minimum and maintain their accreditations. Jane Doe holds a master's degree in Software Engineering and is considered the most tech-savvy person at WPHC. Jane has been constantly upgrading his knowledge about the latest trends in IT and has done much research on the major software solution providers. As the result of her attempts, Jane has created a new project to computerize their paper-based processes. The main goal of this project is to develop a new system to speed up the data entry process. This system needs to digitize a pen-based document at the point of its creation rather than requiring an extra capture step.

Jane has decided to approach Solution@CSCE5430.Projects, Inc. to develop this software system. The system is called iCARE and assumed to run on Windows-based desktop computers and tablet PCs.

System Requirements

After much research and groundwork, Jane has prepared a functional specifications list which details the features that the iCARE, Inc. Web site will have. A copy of this list is shown below.

1. The system shall provide a user friendly interface for the clinician that saves time by eliminating data entry and has maximum flexibility to meet the needs of a highly-complex, regulated and constantly changing environment.
2. Physicians shall have the ability to easily find the clinically important nuggets of information that are so important to quality patient care.
3. Pen based documents are digitized from the point of creation, rather than requiring an extra capture step.
4. Taking advantage of the unique features of the Tablet PC and its "digital ink", the system permits the healthcare worker to be truly mobile by means of pen based computing. This is done by allowing all work tasks to be completed without touching a mouse or keyboard.
5. A fully configurable and easy to deploy software solution

Detail Requirements

After thoroughly analyzing the system's requirements, Jane Doe and Solution@CSCE5430.Projects, Inc. have decided that the iCARE shall provide the following functionalities:

- iCARE shall provide a simple way to log in using a pen only. The user taps his name and can "ink" his password in the box. The user can of course also type (typically using the on-screen keyboard) into the box if desired. Once the user has logged in, the set of controls dynamically changes to match the permissions of this account.
- "iCARE Board" and "My Board" are two main forms to be provided. iCARE Board shall bring up patient lists corresponding to the geographic unit view. Worker will assign himself to patients from the iCARE Board. "My Board" shall show only the active patients list of the logged on person, and hence My Board view can be different for doctors and nurses.
- Palette will be another UI component to help simplify the choosing of documents among large number of potential documents. It's one tablet screen in size so the user can see all choices in one view which consists of a series of buttons for the user to tap on. The Palette shall be able to manage larger number of document choices than can fit on one screen.
- iCARE shall provide a tool to create a new document and to be able to insert different types of text into the new and/or the old documents such as treatments, drug orders and prescriptions. This tool should also be able to recognize generic and brand drug names from a standard reference and supports auto completion.
- iCARE shall provide a tool to import images using a scanner or by browsing the internal stored image files into the application. These imported files will then be converted to PDFs, to become a part of the iCARE documents repository.
- Every PDF document in iCARE should be tagged with a metadata that includes Patient ID, Date of creation/imported, the user ID of the creator/importer, Modification date, the user ID of the modifier, Brief description.
- Only iCARE administrator has the privileges to add workers (doctors and nurses) into the system, edit and delete their records. The administrator account will be shipped with the system.
- Workers of iCARE shall be able to maintain (add, modify) patient records. Patient's record should maintain (but not limited to) the following information: ID, name, address, date of birth, height, weight, bloodGroup, BedID, Treatment area. Each patient record will be associated with zero or more treatment records and a set of digital documents.